

ZEYAD MOHAMED AHMED REFAEY

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PROFESSIONAL SUMMARY

Results-driven Data Science student with hands-on experience in machine learning, deep learning, and NLP. Proven track record of developing and deploying end-to-end ML solutions including sentiment analysis systems, RAG-powered AI assistants, and customer segmentation models. Skilled in Python, TensorFlow, PyTorch, and FastAPI with expertise in building production-ready applications. Completed 100-hour advanced training in Generative AI, Deep Learning, and Building RAG Agents with LLMs through NVIDIA DLI programs. Strong foundation in data analysis, statistical modeling, and interactive dashboard creation.

EDUCATION

Bachelor of Science in Data Science | Egyptian Chinese University (ECU)

Expected Graduation: 2027 | Faculty of Computers and Information Technology

PROFESSIONAL EXPERIENCE

Data Science Intern | Fawry

Cairo, Egypt (Onsite) | December 2025 – January 2026

- Developed an NLP-powered sentiment analysis system using LSTM and GRU architectures in TensorFlow/Keras, deployed via REST API using Flask and FastAPI for real-time predictions
- Performed financial customer segmentation by applying K-Means clustering and Elbow Method to identify 5 distinct behavioral patterns from transactional data
- Built and optimized a feedforward neural network for handwritten digit classification on MNIST dataset, achieving high accuracy through data normalization and categorical cross-entropy optimization
- Engineered predictive models for retail customer spending using Linear Regression with extensive feature engineering and exploratory data analysis
- Conducted comprehensive data cleaning and visualization using Pandas, NumPy, and Seaborn to analyze complex transaction behaviors and generate actionable business insights

TECHNICAL PROJECTS

Smart Contract Summary & Q&A; Assistant | Python, LangChain, RAG, OpenRouter, Gradio

GitHub: github.com/z3yad30/Contract-Summary-Q-A-Assistant

- Built an end-to-end Retrieval-Augmented Generation (RAG) system for legal document analysis, processing PDF/DOC contracts with intelligent chunking and semantic search
- Implemented local vector store using Chroma DB and sentence transformers for embedding generation, enabling accurate document retrieval
- Developed conversational Q&A; interface with inline citations and source tracking using LangChain LCEL chains and AuroraLLM via OpenRouter API
- Created production-ready Gradio web application with document upload, persistent chat history, and one-click summarization features

IMDB Sentiment Analysis API | RoBERTa, FastAPI, Transformers, PyTorch

GitHub: github.com/z3yad30/roberta-large-sentiment-imdb

- Developed production-ready sentiment analysis REST API using fine-tuned RoBERTa-large model, achieving 93-95% accuracy on IMDB test dataset (25,000 reviews)
- Implemented FastAPI backend with Pydantic validation, supporting both single and batch predictions (up to 100 reviews) with confidence scoring
- Built responsive HTML/CSS/JavaScript frontend with real-time analysis, probability distribution visualization, and mobile-optimized interface
- Deployed GPU/CPU agnostic inference pipeline with comprehensive error handling and health monitoring endpoints

Bank Customer Segmentation & Clustering App | K-Means, PCA, Streamlit, scikit-learn

GitHub: github.com/z3yad30/bank-customer-segmentation-clustering

- Performed unsupervised machine learning analysis on 45,211 bank customers to identify 5 distinct behavioral segments for targeted marketing strategies
- Implemented and compared K-Means, DBSCAN, and Hierarchical Clustering algorithms with comprehensive evaluation metrics
- Applied Principal Component Analysis (PCA) for dimensionality reduction and created 2D visualizations of customer segments
- Deployed interactive Streamlit web application enabling real-time customer classification with behavioral interpretation and segment profiling

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **ML/DL Frameworks:** TensorFlow, Keras, PyTorch, scikit-learn, Hugging Face Transformers
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI
- **NLP & LLMs:** LangChain, Sentence Transformers, Prompt Engineering, RAG, RoBERTa, LSTM/GRU
- **Web Frameworks:** FastAPI, Flask, Streamlit, Gradio
- **Databases & Tools:** Chroma DB, Vector Stores, SSIS, Git, OpenRouter API
- **Core Competencies:** Machine Learning, Deep Learning, NLP, Statistical Analysis, Data Visualization, Model Deployment, ETL

TRAINING & PROFESSIONAL DEVELOPMENT

Generative AI Intern - Advanced Level | Information Technology Institute (ITI) - NVIDIA DLI Program *February 2026 | 100 Hours | Remote*

- Completed intensive 100-hour program covering Deep Learning Basics (10 hrs), Building RAG Agents With LLMs (50 hrs), and Capstone Project (40 hrs)
- Gained hands-on experience in developing sophisticated Retrieval-Augmented Generation agents powered by large language models.

Data Analysis Intern | National Telecommunication Institute (NTI)

June – September 2025 | 90 Hours | Remote

- Completed structured program focused on data cleaning, exploratory analysis, and dashboard creation using Python (Pandas, NumPy, Matplotlib), SQL, Power BI, and SSIS

Generative AI Intern - Beginner Level | Information Technology Institute (ITI) - NVIDIA DLI Summer Program

September 2025 | 35 Hours | Remote

- Completed foundational training in Generative AI, LLM development, Prompt Engineering, and Retrieval Augmented Generation (RAG)

CERTIFICATIONS

- Building RAG Agents with LLMs – NVIDIA Deep Learning Institute
- Building LLM Applications With Prompt Engineering – NVIDIA Deep Learning Institute
- Deep Learning With PyTorch: From Concepts to Applications – Mahara Tech
- AI and Machine Learning – Sprints
- Introduction to Deep Learning – Mahara Tech
- AI for All: From Basics to GenAI Practice – NVIDIA Academy